Mohammad Irfan

Graduate Research Assistant
PhD Candidate in Physics
National High Magnetic Filed Laboratory
Florida State University, FL

+1-850 9809175
irfanamu.ik@gmail.com
mirfan2@fsu.edu
Mohammad-Irfan

RESEARCH INTEREST

Strongly Correlated systems, Topological Insulators, Unconventional Superconductivity

Education

- **Florida State University, Tallahassee, USA** *Aug 2021- Dec 2026*
  Doctor of Philosophy in Physics
- **Aligarh Muslim University, Aligarh, India** *Aug 2018 - Aug 2020*
  Master of Science in Physics
- **Jamia Millia Islamia, New Delhi, India** *Aug 2015- July 2018*
  Bachelors of Science in Physics

Professional Experience

- **National High Magnetic Field Laboratory (MagLab)** *May 2022- Present*
  Graduate Research Assistant Tallahassee, FL, USA
  Supervisor: Dr. Laura H. Greene (lhgreene@magnet.fsu.edu)
  – Synthesis and Characterisation (Thermal, electrical and magnetic properties) of superconducting thin films
  – Operate, calibrate, and maintain lab equipment (PPMS, AJA sputtering and evaporation chambers, vacuum systems) and tools; write Standard Operating Procedures (SOPs) and scientific papers; perform user support for external users.
  – Tunneling spectroscopic studies of heavy fermion superconductors and the Kondo Insulators for bulk crystals and the thin films
  – Responsible for leading the lab activities and training and supervising three REU projects and two undergrad theses
- **Department of Physics, Florida State University** *Aug 2021- May 2022*
  Graduate Teaching Assistant Tallahassee, FL, USA
  Course Advisor: Dr. Vandana Tripathi (vtripathi@fsu.edu)
  – Taught 6 sections of lab courses; was responsible for setting up the experiments, explaining and helping students in writing reports
  – Performed grading duties and assisted the major professor in carrying out teaching realted duties.

Research Projects

- **Planar Tunneling Spectroscopic(PTS) studies of 1-1-5 Heavy Fermion Superconductors**
  Ph.D. Project | Advisor: Prof. Laura H. Greene, Florida State University, Tallahassee, Florida May 2022 - Present
  – Tunnel junction fabrication and Quantum device to understand the cooper pairing in heavy fermion superconductors
  – Tools Used: Vacuum techniques, Ion beam etching, sputtering

- **Planar Tunneling Spectroscopy of possible Topological Kondo Insulator SmB_6 and YbB_{12}**
  Ph.D. Project | Advisor: Prof. Laura H. Greene, Florida State University, Tallahassee, Florida Sep 2023 - Present
  – Probing the topological surface states in Kondo insulator to understand the interaction with excitations
  – Tools used: He-3 cryogenics, Thin film deposition, XRD, AFM

- **Characterization of a Freshly Prepared Ferroelectric Liquid Crystal Mixture**
  MS Thesis | Advisor: Dr. Shikha Chauhan, Aligarh Muslim University, Aligarh, India Oct 2019- Apr 2020
  – Experimental study of dielectric parameters on a newly prepared Ferroelectric Liquid Crystal(FLC) mixture, namely, W302A in the frequency regime 20Hz-2MHz.
  – Techniques Used: LCR spectroscopy, Optical Microscopy

- **Monte Carlo Simulation of the 2-D Ising Model**
  BS Thesis | Advisor: Dr. Rashid Ali, Jamia Millia Islamia, New Delhi, India Jan 2018- Apr 2018
  – Simulation of Ising model to explain the transition of materials from Ferromagnetism to Paramagnetism using Monte Carlo Methods
  – Program Used: C++, Fortran

Skills Summary

Material Synthesis: Thin Film Synthesis, Thermal Vapor Deposition, Sputtering, and Hetro-structure Fabrication
Sample Preparation: Crystal Polishing (angstrom smoothness), Ion Beam Etching, Plasma Oxidation, Sputtering, Thermal Vapour deposition
Instrument Training: AJA thin Film deposition system, Physical Property Measurement System (PPMS), Magnetic Properties Measurement System (MPMS), Thermal evaporator, Vacuum chambers, and He-3 cryogenics
Software: Python, LaTeX, Origin Pro, Auto Desk Inventor, LabVIEW, Matlab
Safety Training:Compressed Gas, Cryogenics, Electrical Safety, Hazard Communication, Hazardous Waste and Magnetic Field Safety

Honors/Awards/Achievements

1. Awarded **ICAM Travel Grant** by theICAM - I2CAM Institute for Complex Adaptive Matter, to support conference travel to the ICAM annual Conference 2026, Davis Caliofornia. (Sep 2026)
2. Won the prestigious **3-min theis presentation award** held at Department of Physics, Florida State University, USA (Apr 2026)
3. Awarded **COGS Presentation Grant** by the Congress of Graduate Students, Florida State University, to support conference travel to the APS Global Summit (Feb 2026).
4. Awarded **Evelyn and John Baugh Research Presentation Scholarship** by the Department of Physics, Florida State University, to support conference travel to the APS Global Summit (Feb 2026).
5. Awarded the 2^{nd} prize for the best poster "Unconventional Superconductivity and the Cooper pair formation in CeCoIn5" at the Annual Winter School and Workshop,Rice Centre for Quantum Materials, Houston, TX (Nov 2025)
6. Awarded the **Yung Li Wang Award** by the Department of Physics, Florida State University for the elucidation in the Topological Kondo Insulator and the Unconventional Superconductivity (Apr 2025).
7. Awarded the **Distinguished Student Award (DS) 2025** by Forum of Early Career Scientists (FECS) and the Forum on International Physics (FIP), American Physical Society (Mar 2025).
8. Awarded the **FSU Quantum Conference Travel Support Award** by the FSU Quantum Mission (Jan 2025, Jan 2026).
9. Awarded the **David L. Wilder Award** by the Department of Physics, Florida State University for the elucidation in the Topological Kondo Insulator and the Unconventional Superconductivity (Aug 2024).
10. Awarded **ICAM Travel Grant** to present the research on Strongly Correlated Systems at the Forschenzentrum Julich (Sep 2024)
11. Qualified **Graduate Aptitude Test Examination(Gate 2020)**, to be eligible to pursue Ph.D. in the topmost research Institutes in India including the IITs.
12. Awarded **Sir Syed Global Scholar Award (2019)** by the Sir Syed Education Society of North America, which recognizes emerging scholars from India.
13. Awarded 1^{st} Prize in '**National Science Day Celebration-2019**' held at the Department of Physics, A.M.U, Aligarh, India.
14. Secured 2^{nd} position in my undergraduate studies at Department of Physics, Jamia Millia Islamia (2018).

Positions of Responsibility

- **President,** American Physical Society, Florida State University Student Chapters *May 2025- Dec 2025*
- **Vice- President,** American Physical Society, Florida State University Student Chapters *2024-2025*
- **Lab Safety Officer,** Greene's Lab, National High Magnetic Field Lab *2023- 2026*
- **Vice President,** Cricket Association at Florida State University *2024-2025*

- **Treasurer,** Physics Graduate Student Association, Florida State University *2024-2025*
- **Application Team Lead,** Sir Syed Society of North America (www.ssgsa.us) *2023-2024*
- **Secretary,** American Physical Society, Florida State University Student Chapters *2023-2024*
- **REU co-mentor,** National High Magnetic Field Laboratory *Summer 2023, 2025, 2026*
- **Finance Lead,** Sir Syed Society of North America (www.ssgsa.us) *2022- 2025*
- **Organising Team Member,** Cultural Committee, F.R.K Hostel, Jamia Millia Islamia *2017-2018*
- **Student Representative,** Subject Association, Department of Physics, Jamia Millia Islamia *2017-2018*

Selected Oral/ Poster presentation

1. Planar Tunneling Spectroscopy of SmB6 thin films (Poster) *Florida Quantum Conference, 2025* (Apr 09, 2026)
2. Planar Tunneling Spectroscopy of SmB6 thin films (Oral) *PS Global Summit March 19, 2026 (Colorado)*
3. Development of MgO tunnel barrier to probe the superconductivity in 1-1-5 heavy fermions (Poster, 2nd prize) *Winter School on Material Synthesis, RICE University, Houston* (Nov 2025)
4. Development of MgO tunnel barrier to probe the superconductivity in 1-1-5 heavy fermions (Poster) *Annual Meeting ASEMFL 2025, Orlando*
5. Development of MgO tunnel barrier to probe the superconductivity in 1-1-5 heavy fermions (Oral) *APS Global Summit 2025, Anaheim, CA, USA (March 2025)*
6. Development of MgO tunnel barrier to probe the superconductivity in 1-1-5 heavy fermions (Poster) *Dirac Quantum Discussion, National High Magnetic Field Lab, Tallahassee, FL, USA (Feb 2025)*
7. Probing the surface states of Kondo insulator YbB12 using second harmonic measurements (Poster) *Autumn School on Correlated Systems, Julich , Germany (Sep 2024)*
8. Tunneling spectroscopic study of the Kondo insulator YbB12 using second harmonic measurements (Oral) *APS March Meeting 2024, Minneapolis, USA (Mar 2024)*
9. Probing the Surface State of SmB6 through Planar Tunneling Using 2nd Harmonic Detection Techniques (Oral) *APS March Meeting 2024, Minneapolis, USA (Mar 2024)*
10. Robust Topological Surface States in YbB12: A Comparative Study with SmB6 using planar tunneling spectroscopy (Oral) *NanoFlorida International Conference, Tallahassee, FL, USA (Apr 2024)*
11. Optimization of parameters or Nb Thin films and MgO Barrier for Planar Tunnel Junctions (Poster) *Department of Physics Poster Competition 2024, Tallahassee, FL, USA (Apr 2024)*
12. Dielectric properties of a newly prepared ferroelectric Liquid crystal mixture (oral) *International Conference on Recent Advances in Engineering and Science, A.M.U Aligarh, India. (Jan 2020)*

Short term Courses/ Professional Development

1. Attended the Workshop on ***Neutron Scattering, High Magnetic Field, and Complimentary Techniques for Quantum Materials*** by the FSU Quantum Initiative, Tallahassee, FL (Feb 2025).
2. Attended the Winter School on ***Quantum Material Synthesis*** and Workshop on ***Quantum Geometry*** organized by the Rice Centre for Quantum Materials, Houston (Nov 2025).
3. Attended the Theory Winter School ***Strong Correlation Across Newly Accessible Length and Energy Scales*** organized by the National High Magnetic Field Lab (Jan 2025).
4. Attended the Autumn school on ***Correlations and Phase Transition*** organized by the Institute for advanced Simulation at Forschungszentrum Julich (Sep 2024).
5. Attended the ***Dirac Quantum Discussion*** organized by FSU Quantum Initiative for the future research in Quantum materials and Quantum computation platforms. (2023, 2024, 2025, 2026).

6. Attended and represented the APS FSU Chapters at ***Annual Leadership Meeting*** organized by the American Physical Society (Jan 2024).
7. Attended the short term course on ***Current Trends In Condensed Matter Physics*** organised by the Department of Physics, National Institute of Technology, Jalandhar (Sep 2020).
8. Participated in International webinar on ***Machine Learning in Physics 2020***, conducted by West Bengal State University, India.

Services/ Membership

- **Judge,** Capital Regional Science and Engineering Fair,Tallahassee, FL *Feb, 2026*
- **Quantum Ambassador,** Florida State University Quantum Initiative, Tallahassee, FL *2025-2026*
- **Volunteer,** Physics Open House, Department of Physics, Florida State University *2025*
- **Volunteer,** MagLab Open House, National High Magnetic Field Laboratory *2023, 2025*
- **Mentor,** Sir Syed Global Scholar Award, Sir Syed Society of North America *2023- Present.*